

Satellite Orbital Trace



Ground Positions



Time M: {sat pos, sat vel}

Time N: {sat pos, sat vel}

Pair-i: {UE pos, gNB pos, regen}

Pair-j: {UE pos, gNB pos, trans}

Centralized Ephemeris Engine

Config Loader

NTN Calculator

SHM Writer

NTN Args {
prop delay; doppler shift;
timing advance; k offset; ...
}

	Pair i	Pair j
Time M	Pair-i { # Time M NTN Args { ... } }	Pair-j { # Time M NTN Args { ... } }
Time N	Pair-i { # Time N NTN Args { ... } }	Pair-j { # Time N NTN Args { ... } }

SHM Ring Buffer

1 2 3 4 5 6 7 ... N-1 N

Mem Read Pair 1

Mem Read Pair 2

Mem Read Pair K

Pair 1

Unchanged
3GPP NTN

RRC/NTN-Specific

F1

PDCP/RLC/MAC

nFAPC

3GPP NTN
State
Machine

PHY-to-MAC
Indications

Light-weight
PHY Model

PHY-Lite

Pair 2

Unchanged
3GPP NTN

RRC/NTN-Specific

F1

PDCP/RLC/MAC

nFAPC

3GPP NTN
State
Machine

PHY-to-MAC
Indications

Light-weight
PHY Model

PHY-Lite

...

Pair K

Unchanged
3GPP NTN

RRC/NTN-Specific

F1

PDCP/RLC/MAC

nFAPC

3GPP NTN
State
Machine

PHY-to-MAC
Indications

Light-weight
PHY Model

PHY-Lite